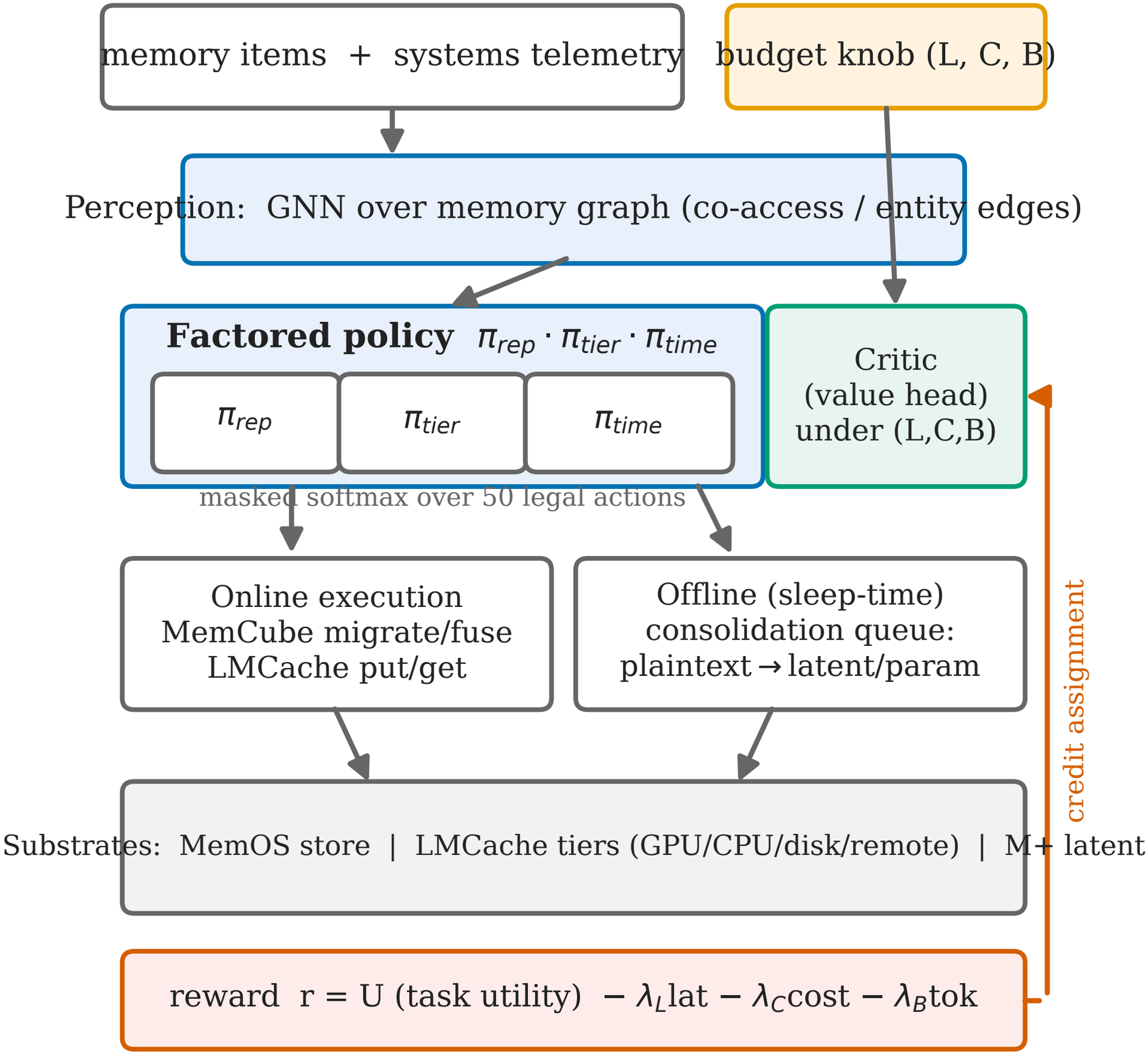
memory items + systems telemetry
budget knob (L, C, B)
Perception: GNN over memory graph (co-access / entity edges)
Factored policy $\pi_{rep} \cdot \pi_{tier} \cdot \pi_{time}$
π_{rep}
π_{tier}
π_{time}
Critic (value head) under (L,C,B)
masked softmax over 50 legal actions
Online execution MemCube migrate/fuse LMCache put/get
Offline (sleep-time) consolidation queue: plaintext → latent/param
Substrates: MemOS store | LMCache tiers (GPU/CPU/disk/remote) | M+ latent
reward r = U (task utility) $-\lambda_L$lat $-\lambda_C$cost $-\lambda_B$tok
credit assignment